Assignment 6: SQL Aggregation, Filtering & Conditional Classification

Overview:

In this assignment, you will create an orders table, insert sample data, and write SQL queries to aggregate sales, filter results, and classify records using conditional logic.

1. Create and Populate Table

```sql
CREATE TABLE Orders (
  OrderID INT PRIMARY KEY,
  CustomerName VARCHAR(50),
  ProductCategory VARCHAR(50),
  Quantity INT,
  TotalPrice DECIMAL(10,2),
  OrderDate DATE
);

INSERT INTO Orders (OrderID, CustomerName, ProductCategory, Quantity, TotalPrice, OrderDate) VALUES
  (1, 'Alice',   'Electronics', 2,  1600.00, '2024-11-01'),
  (2, 'Bob',     'Furniture',   1,   300.00, '2024-11-02'),
  (3, 'Charlie', 'Electronics', 1,   800.00, '2024-11-03'),
  (4, 'Diana',   'Stationery', 10,           50.00, '2024-11-04'),
  (5, 'Eve',     'Electronics', 3,  2400.00, '2024-11-05'),
  (6, 'Frank',   'Stationery', 20,   100.00, '2024-11-06');
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1)** and save it in the same **PDF/Doc/Word** file.

2. SQL Query Tasks

1. **a) Total Quantity by Category:** Compute total Quantity sold for each ProductCategory.

SQL:

```
SELECT ProductCategory, SUM(Quantity) AS TotalQuantity
FROM Orders
GROUP BY ProductCategory;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2a)*** *and save it in the same* ***PDF/Doc/Word*** *file*

2. **b) Categories with High Sales Volume:** Find categories where total Quantity > 10.

SQL:

```
SELECT ProductCategory
FROM (
  SELECT ProductCategory, SUM(Quantity) AS TotalQuantity
  FROM Orders
  GROUP BY ProductCategory
) AS T
WHERE TotalQuantity > 10;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2b)*** *and save it in the same* ***PDF/Doc/Word*** *file*

3. **c) Average Order Value by Category:** Calculate average TotalPrice per ProductCategory.

SQL:

```
SELECT ProductCategory, AVG(TotalPrice) AS AvgPrice
FROM Orders
GROUP BY ProductCategory;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2c)*** *and save it in the same* ***PDF/Doc/Word*** *file*

4. **d) Categories with Low Average Price:** Identify categories where average TotalPrice < 1000.

SQL:

```
SELECT ProductCategory
FROM (
  SELECT ProductCategory, AVG(TotalPrice) AS AvgPrice
  FROM Orders
  GROUP BY ProductCategory
) AS T
WHERE AvgPrice < 1000;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2d)*** *and save it in the same* ***PDF/Doc/Word*** *file*

5. **e) Classify Orders by Value:** Add a column marking each order as 'High Value' if TotalPrice > 1000, else 'Low Value'.

SQL:

```
SELECT *,
  CASE WHEN TotalPrice > 1000 THEN 'High Value'
       ELSE 'Low Value'
```

```
  END AS ValueCategory
FROM Orders;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2e)*** *and save it in the same* ***PDF/Doc/Word*** *file*

> Tip: If you see "table already exists", rename your table (e.g., Orders1) before running the CREATE statement.

Submission Guidelines:

Create One Main Folder (named Assignment6_YourName) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment6_YourName.sql**

Convert the folder into ZIP format.
Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.